

Bryan Philinathaniel Hutagalung

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A driven Information Systems and Technology student at Bandung Institute of Technology with a strong foundation in data science, AI, and machine learning. Experienced in developing and deploying machine learning models to solve real-world problems, with a focus on optimizing performance and scalability. Proven leadership in managing teams and delivering data-driven solutions, backed by certifications in AI and data science from IBM and Google.

Educations

Bandung Institute of Technology | Bandung, Indonesia

August 2022 – August 2026 (Expected)

- Bachelor of Science in Information System and Technology. GPA: 3.54/4.00
- Related Courses: Algorithm and Data Structure, Algorithm Strategy, Object Oriented Programming, Database Modelling, Database Management, Advanced Artificial Intelligence, Machine Learning, Data Mining, Data Management

Work Experiences

PT Global Niaga Sakti – Backend Engineer (Contract, Remote) | Bandung, Indonesia

April 2026 – May 2026

- Built an integrated internal platform automating the full operational and financial workflow, covering quotation, purchase order, delivery note, invoice, tax reporting, and accounting in a single system.
- Reduced processing time by 50%+ compared to the previous manual workflow by consolidating previously siloed processes across teams.

PT Bank Neo Commerce Tbk. – Data Scientist (Intern) | Jakarta, Indonesia

June 2025 – September 2025

- Drove a 90%+ reduction in model deployment time (from over a month to under 3 days) by engineering a fully automated MLOps pipeline on Kubernetes using Kubeflow, MLflow, and KServe.
- Built a centralized Feature Store using Feast, eliminating data redundancy across teams while serving production features for real-time inference with a verifiable P99 latency of less than 50ms.
- Standardized the ML workflow with a modular microservices architecture, enabling new models to be deployed via configuration changes, which minimized manual engineering and ensured end-to-end reproducibility.

PT Global Niaga Sakti – Frontend Engineer (Contract, Remote) | Bandung, Indonesia

January 2025 – March 2025

- Redeveloped the company's multilingual corporate website using Astro, Tailwind CSS, and TypeScript, increasing site performance by 300% with a 95+ Google Lighthouse score over the previous WordPress implementation.
- Built an internationalization system supporting English and Indonesian, resulting in a 40% increase in international client engagement and reducing bounce rates by 25% for non-Indonesian visitors.
- Improved organic search visibility by 65% and increased lead generation through contact forms by 50% within the first three months through optimized SEO and accessible UI implementation.

Organizational Experiences

HMIF ITB – Chairman | Bandung, Indonesia

May 2025 – March 2026

- Led 300+ staff and 200 interns across the HMIF ITB executive board, overseeing operations for 1,000+ active members.
- Coordinated 50+ events with 1,000+ total participants, including cross-institutional collaboration with IAIF ITB and the STEI ITB department.
- Improved administrative efficiency by ~50% and grew organizational revenue 3x through restructuring internal management systems.

HMIF ITB – Deputy Head of Internal Department | Bandung, Indonesia

May 2024 – May 2025

- Organized and oversaw 4+ large-scale events with 1,500+ participants, achieving a 60% success rate in meeting organizational targets during the first half of my tenure.
- Led 65 members across 3 divisions, ensuring smooth operations and fostering a supportive environment for over 1,000 HMIF ITB members.

Kabinet KM ITB – General Director of Research | Bandung, Indonesia

September 2023 – April 2024

- Produced 3 research documents from data collected across 44 Himpunan Mahasiswa Jurusan (HMJ), informing the evaluation of the cadet system across each HMJ and KM ITB.
- Led a team of 3 researchers, delivering insights that drove academic and organizational improvements across ITB student organizations.

Achievements

Find IT! 2025 – Finalist Data Analyst Competition | Yogyakarta, Indonesia **April 2025**

- Developed a machine learning model to predict COPPA compliance risks in mobile apps by analyzing app metadata to identify child-targeting signals, reducing false negatives in risk detection.
- Built an ensemble model (XGBoost, CatBoost, LightGBM, Random Forest) achieving 97% recall for high-risk apps, directly improving compliance screening accuracy.
- Created an end-to-end ML pipeline that reduced manual compliance review by 60% while maintaining detection accuracy, with clear output through automated reports and visualizations.

Volunteering

ARKAVIDIA 9.0 – Head of Data Science Academy Sub-Division | Bandung, Indonesia **October 2024 – February 2025**

- Developed a Data Science curriculum with 4 milestones spanning introduction to storytelling, validated with industry experts, faculty, and current standards.
- Designed the selection process, 4 hands-on projects, and 1 capstone project, ensuring participants gained practical and measurable experience.
- Recruited 15+ industry speakers, mentors, and judges, giving academy participants direct exposure to real-world Data Science applications.

Perayaan Wisuda Oktober HMIF ITB 2024 – Project Officer | Bandung, Indonesia **July 2024 – October 2024**

- Organized a 2-month preparation and 3-day main event running concurrently with OSKM ITB 2024, achieving a 90% satisfaction rate among 1,000+ attendees.
- Led 306 committee members across 6 departments to ensure smooth execution and high participant engagement.

OSKM ITB 2024 – Field Coordinator | Bandung, Indonesia **July 2024 – September 2024**

- Coordinated a 2-month preparation and 5-day main event running full days (dawn to night), managing 1,151 field members and logistics for 7,500+ participants while reducing wait time by 30%.
- Delivered 5 educational sessions to new students, covering safety procedures and campus orientation.

Projects

Scratch NN for Modeling and Predicting | Python, ANN, FFNN, CNN, LSTM, RNN | Repository 1 | Repository 2 **March 2025**

- Built a neural network framework from scratch with 7+ activation functions, 5 weight initialization methods, and multiple optimization techniques, achieving >93% accuracy on MNIST digit classification and reducing model optimization time by 40%.
- Implemented Adam optimization, batch normalization, and gradient clipping in a modular ML library, demonstrating 35% faster convergence than baseline and enabling systematic comparison across architectures and hyperparameters.

Chatbot for Document Reader | LangChain & Transformers, PyTorch, FAISS, Streamlit | Repository **January 2025**

- Achieved 70% faster document processing via GPU acceleration and efficient memory management, enabling real-time analysis of large PDFs with response times under 2 seconds.
- Processed and analyzed PDF documents with 85% query accuracy while supporting concurrent users and maintaining GPU memory usage below 60%.

Scratch Algorithm for Modeling and Predicting | Python, AI, ML, KNN, GNB | Repository **December 2024**

- Built custom ML algorithms achieving 94.7% F1-score for phishing detection, on par with industry-standard implementations.
- Processed 98,000+ data points using an optimized KD-Tree implementation, reducing prediction time on high-dimensional data.

Scratch Algorithm for Magic Cube Solver | Python, React.js, RESTful API, Tailwind CSS | Repository **November 2024**

- Implemented 6 local search algorithms (Steepest Ascent, Sideways Move, Stochastic, Random Restart, Simulated Annealing, Genetic Algorithm), achieving 85% faster solution convergence than traditional methods.
- Built an interactive 3D visualization with real-time playback at 60 FPS, handling 1,000+ cube states and increasing user engagement by 90%.

Certifications

- **IBM AI Engineering** | Python, PyTorch, Tensorflow, Transformers, Gradio, LLM, LangChain | Credentials
- **IBM Data Science** | Python, SQL, Scikit-Learn, GenerativeAI | Credentials
- **Google Data Analytics** | R, SQL, Tableau, Spreadsheet | Credentials